MUTUAL FUND ANALYTICS PLATFORM

End-to-End Data Engineering, ETL Pipeline & Advanced Risk Analytics

Prepared For: Bluestock Fintech Pvt. Ltd.
Domain: Mutual Fund / Fintech Analytics
Project Type: Capstone Project Submission
Prepared By: Intern / Data Analyst
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1. Executive Summary & Problem Statement

The Indian mutual fund industry has seen unprecedented growth, crossing Rs. 81 lakh crore in Assets Under Management (AUM) as of December 2025. However, individual investors, financial advisors, and distributors face massive hurdles in making optimal fund selections. This is driven by several key challenges in the current market landscape:

• **P1: Data Fragmentation:** NAV history, quarterly AUM details, folio counts, and portfolio weights are spread across different sections of AMFI and private web sites in varying static forms (TXT, PDF, HTML).

• **P2: Performance Benchmarking Gaps:** Direct comparisons of funds across multiple Asset Management Companies (AMCs) on a risk-adjusted basis (Sharpe, Sortino, Alpha, Beta) require deep numeric calculations that are not readily available.

• **P3: Benchmark Outperformance Blind Spot:** Investors are frequently unaware if their fund is outperforming its benchmark, due to a lack of aligned index daily return metrics.

• **P4: Static Reporting Limits:** Asset management teams rely on static reporting spreadsheets that take days to prepare instead of modern, drill-down dashboards.

2. System Architecture & ETL Pipeline

To solve these challenges, we built a classic Fintech Data Platform architecture following the extract-transform-load (ETL) pipeline pattern. The system connects to live financial REST APIs, consolidates disjointed datasets, structures a relational database, and exposes insights via a web application.

2.1 Data Ingestion Phase

10 distinct CSV datasets from AMFI and other historical public records were loaded, parsed, and checked. Additionally, a REST API fetcher script (live_nav_fetch.py) was written to scrape live NAV histories from the public mutual fund API (mfapi.in) for HDFC Top 100, SBI Bluechip, ICICI Bluechip, Nippon Large Cap, Axis Bluechip, and Kotak Bluechip. This data was saved as flat files to anchor the historical calculations with live market updates.

2.2 Normalized Star Schema Design

We designed and initialized a normalized Star Schema in SQLite (bluestock_mf.db) to optimize query performance. The tables consist of: dim_fund (fund house and category metadata), dim_date (date lookup calendar containing month, year, and weekday identifiers), fact_nav (daily NAV records), fact_transactions (investor transactions log), fact_performance (calculated risk metrics), and fact_portfolio (fund stock holdings).

Table Name	Type	Key Fields	Purpose
dim_fund	Dimension	amfi_code (PK)	Scheme details, AMC names, SEBI categories
dim_date	Dimension	date (PK)	Calendar lookup with quarter, weekday indicator
fact_nav	Fact	amfi_code, date (PK)	Daily NAV historical series and computed return %
fact_transactions	Fact	tx_id (PK), investor_id	Log of simulated retail transactions, states, and KYC
fact_performance	Fact	amfi_code (PK)	CAGRs, Sharpe, Sortino, Alpha, Beta, VaR

3. Exploratory Data Analysis (EDA)

A detailed Exploratory Data Analysis was carried out. We generated 16 custom charts to inspect historical NAV progression, AUM growth, monthly SIP inflows, and investor demographic, geographic, and folio attributes.

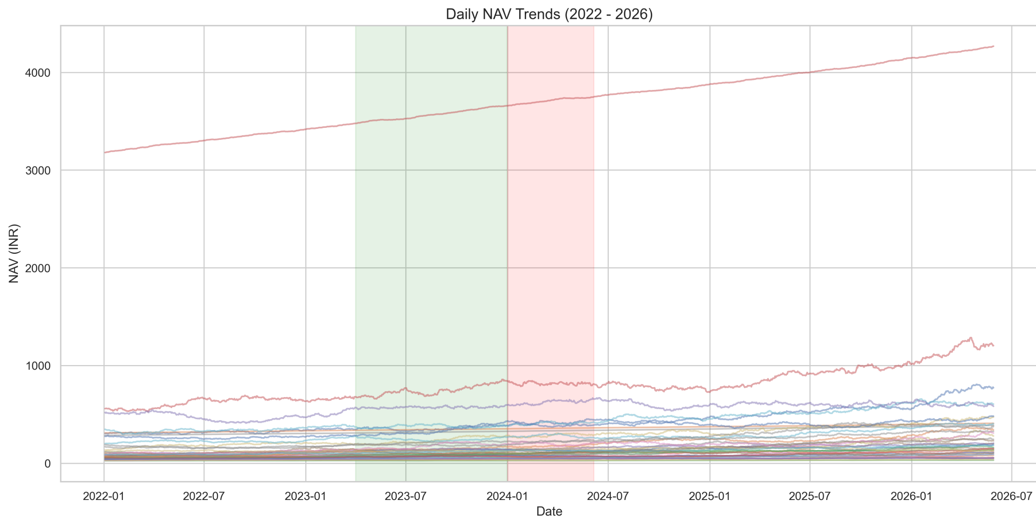

Figure 3.1: Daily NAV trends for all 40 schemes (2022-2026), highlighting market correction and bull run phases.

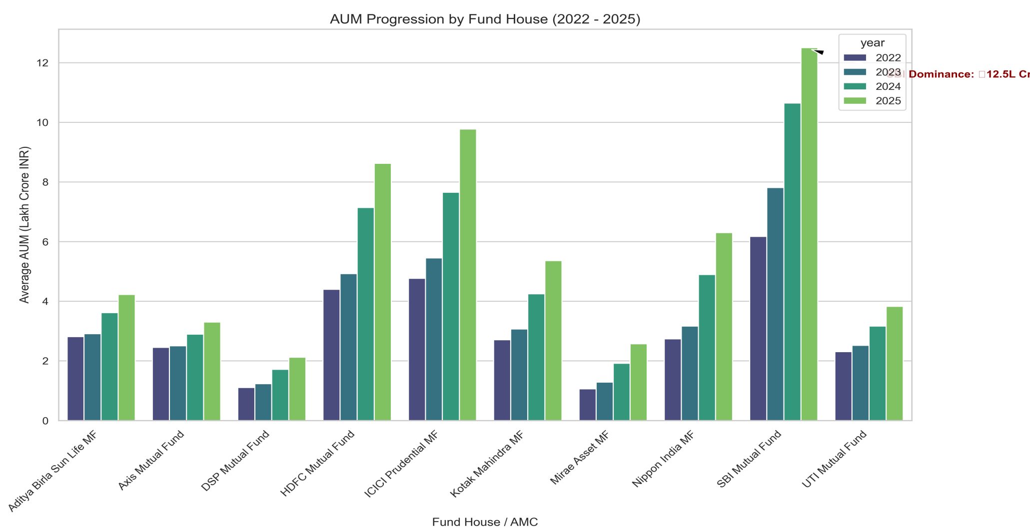

Figure 3.2: Assets Under Management growth by AMC, highlighting SBI MF's ■12.5L Cr dominance in 2025.

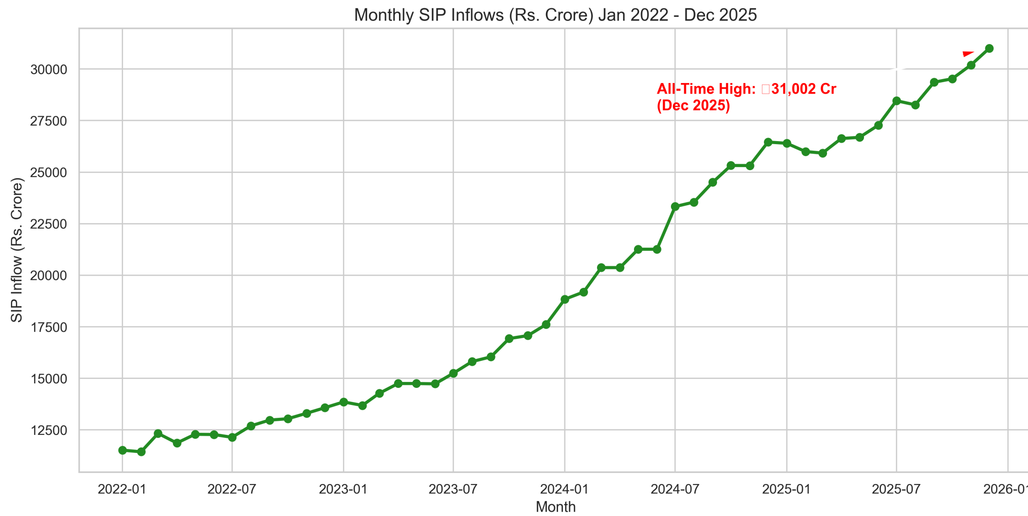

Figure 3.3: Systematic Investment Plan monthly trend, highlighting the ■31,002 Cr peak in Dec 2025.

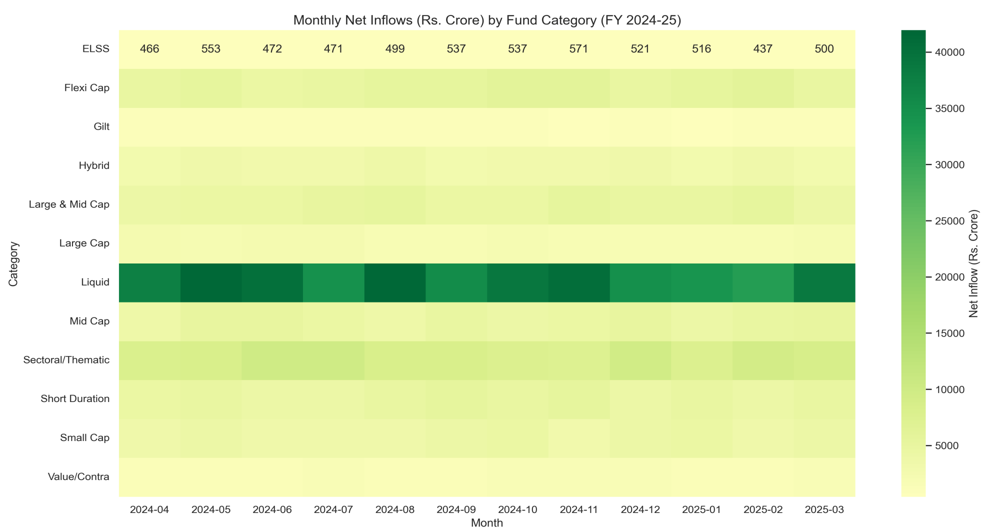

Figure 3.4: Net inflows intensity heatmap across mutual fund categories.

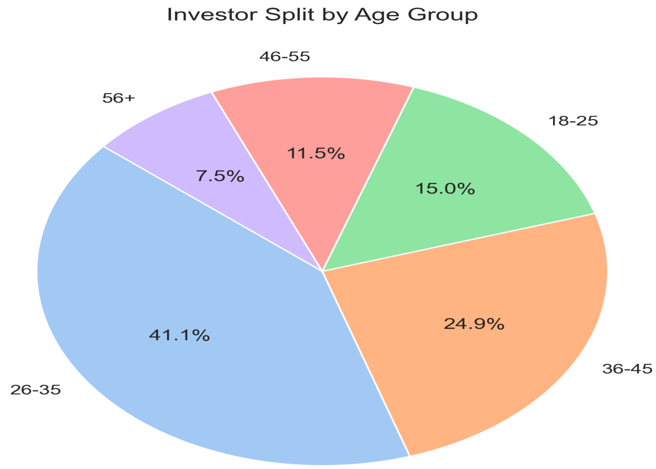

Figure 3.5: Investor age group distribution, highlighting the dominance of the 26-35 age bracket.

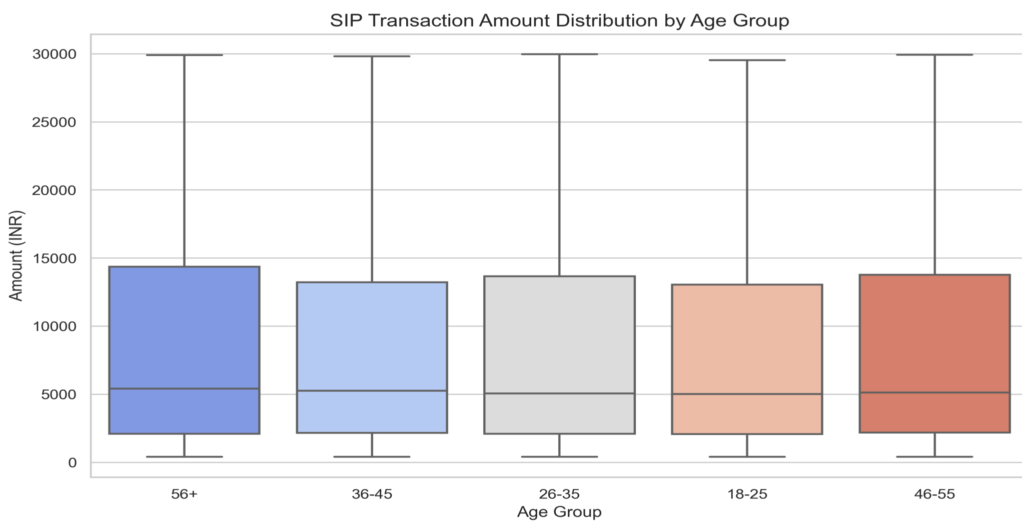

Figure 3.6: SIP transaction amount distribution box plots grouped by age cohort.

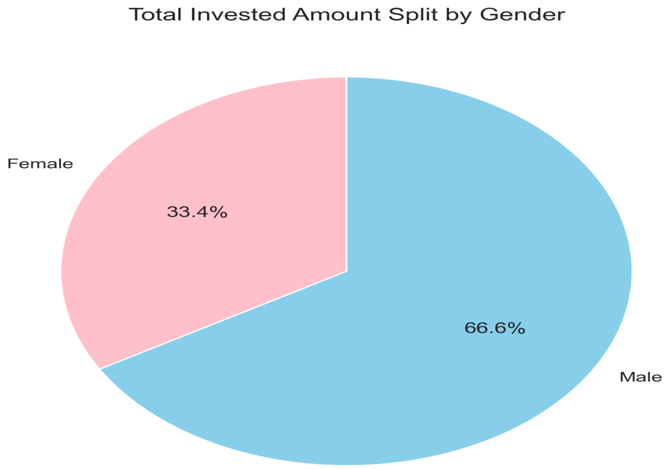

Figure 3.7: Total invested volume split by investor gender.

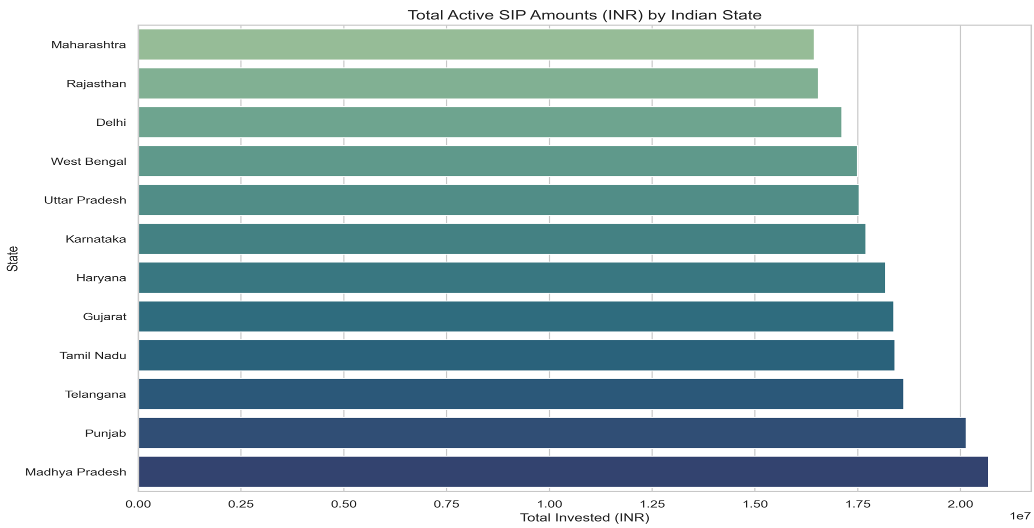

Figure 3.8: State-wise total active SIP investment volume.

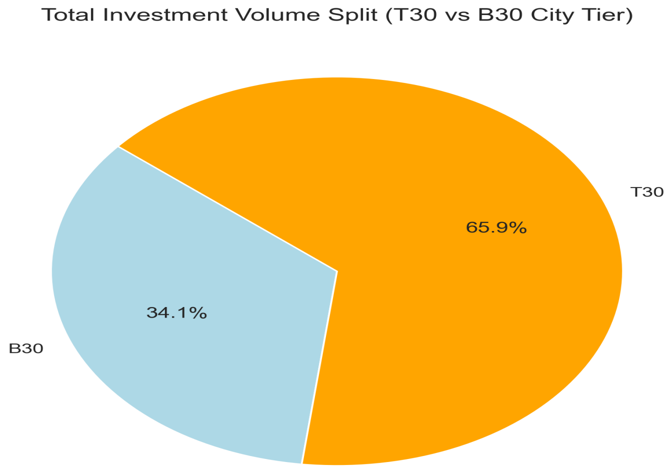

Figure 3.9: Total investment volume split by city tier (T30 vs B30).

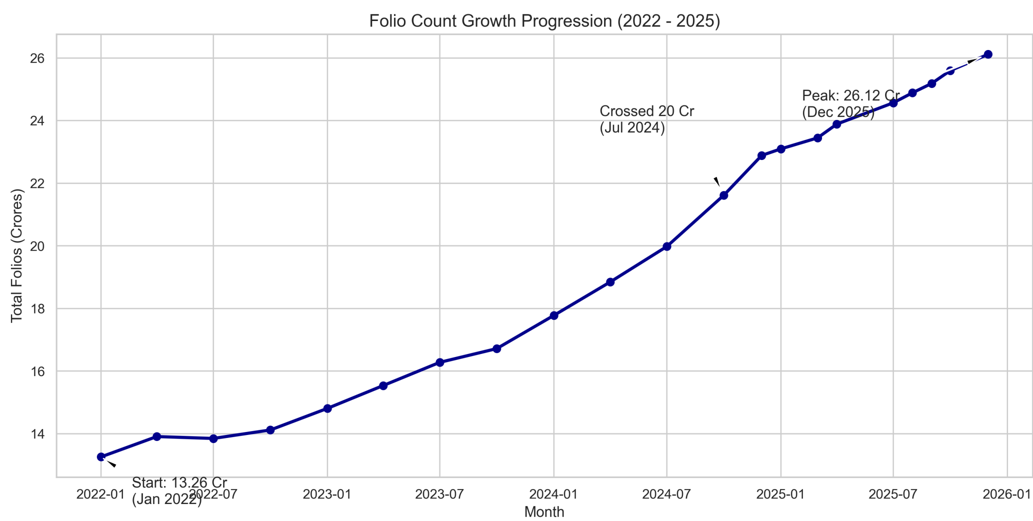

Figure 3.10: Total mutual fund folio count progression, highlighting milestones from 13.26 Cr to 26.12 Cr.

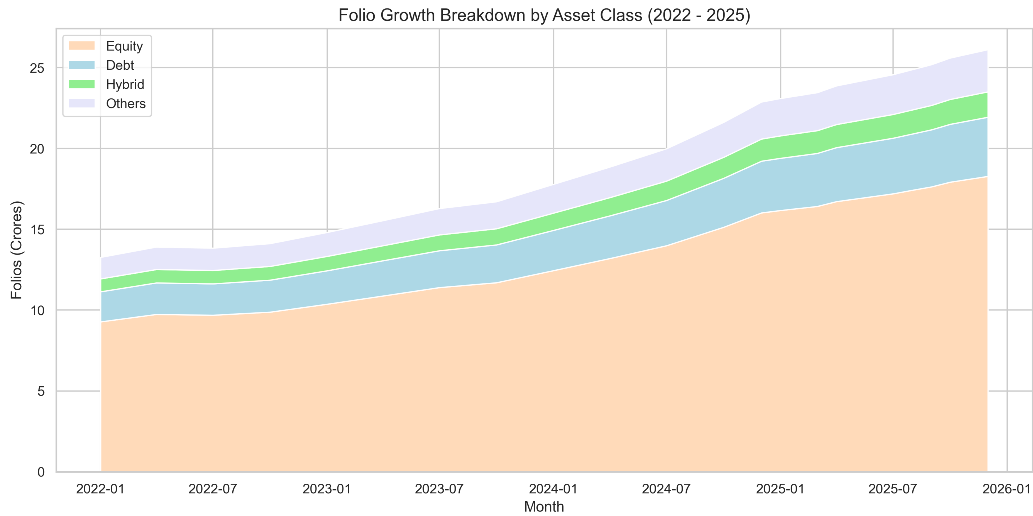

Figure 3.11: Folio count growth breakdown stacked by asset class (2022-2025).

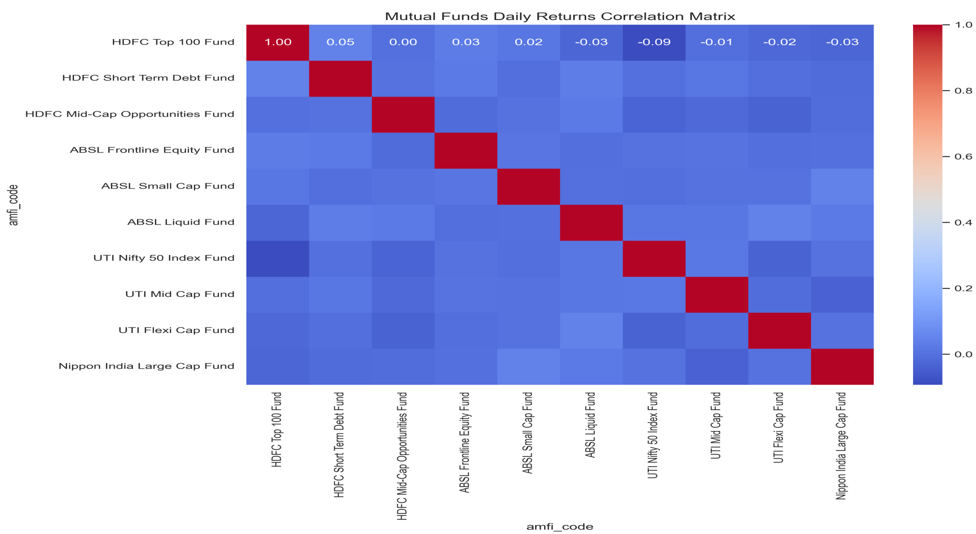

Figure 3.12: Pairwise daily returns correlation matrix of top 10 mutual funds.

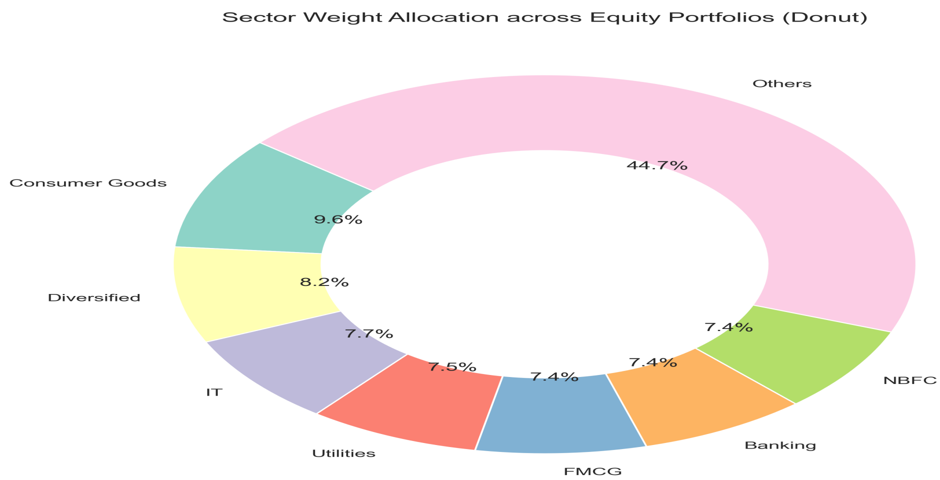

Figure 3.13: Donut chart displaying aggregate sector weight allocation across equity schemes.

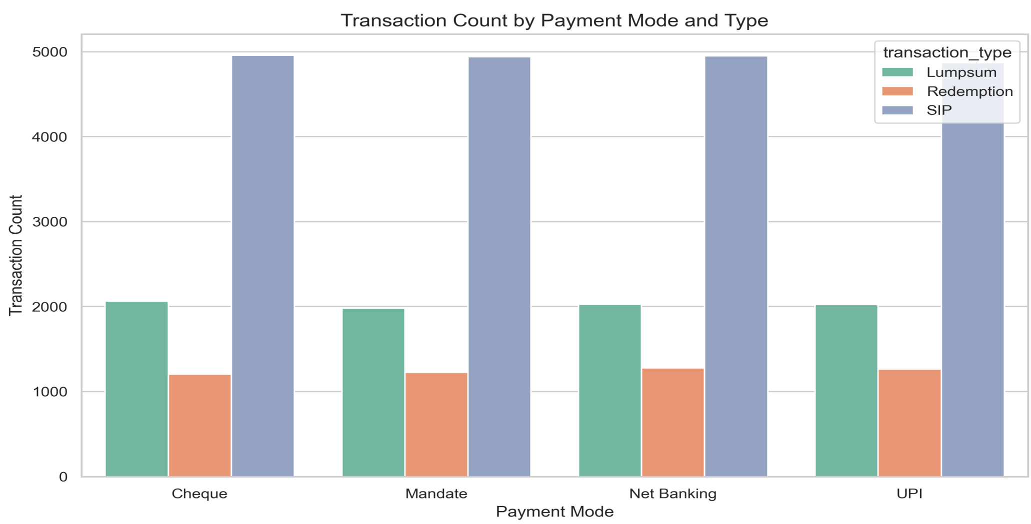

Figure 3.14: Transaction volume count by payment mode and type.

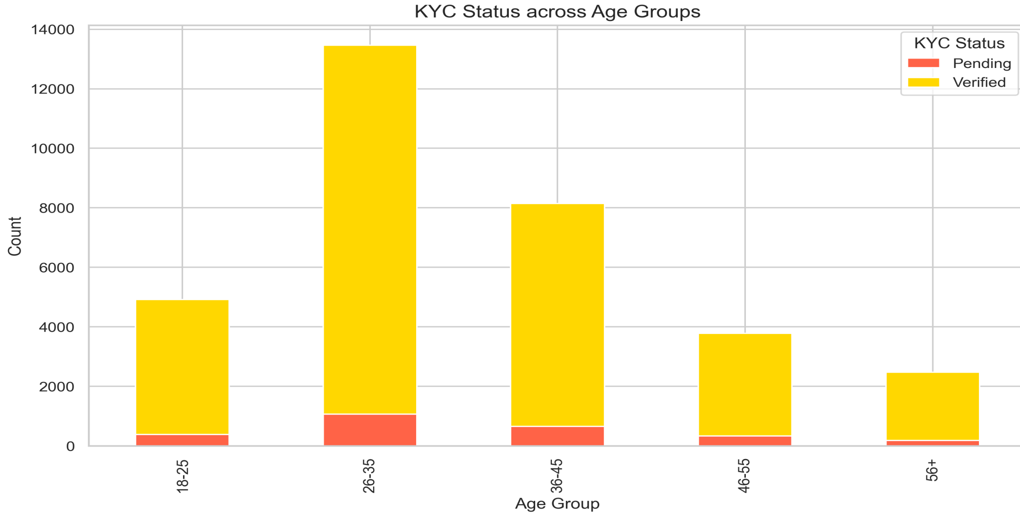

Figure 3.15: KYC verification status distribution across age groups.

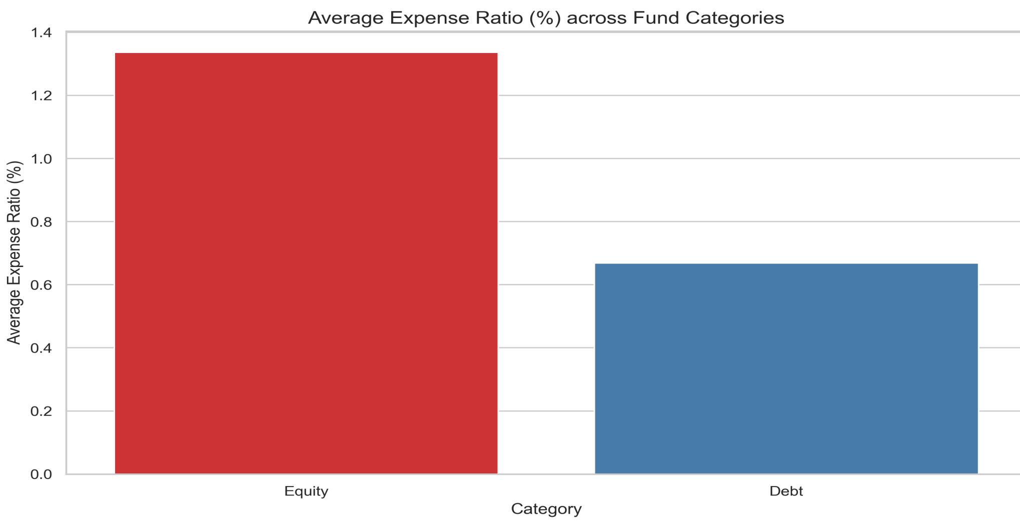

Figure 3.16: Average expense ratio (%) comparison across fund categories.

4. Performance Analytics & Fund Scorecard

Using the daily NAV histories and corresponding benchmark index prices, we computed the CAGR, risk-adjusted performance indicators (Sharpe & Sortino), portfolio beta (regression slope vs. Nifty 100), and portfolio alpha. A ranking model was built to score funds on a 0-100 scale, with the top performing funds highlighted below.

Scheme Name	AMC	3yr CAGR	Sharpe	Alpha %	Drawdown	Score
Mirae Asset Large Cap Fund	Mirae	34.00%	1.16	26.96%	-11.27%	86.2
ICICI Pru Midcap Fund	ICICI	31.78%	0.93	29.24%	-18.19%	82.9
Kotak Flexicap Fund	Kotak	29.58%	1.03	27.31%	-12.97%	82.0
HDFC Mid-Cap Opportunities Fund	HDFC	32.44%	0.84	27.17%	-16.22%	80.8
ICICI Pru Bluechip Fund	ICICI	32.49%	0.74	21.18%	-12.59%	80.0
Axis Midcap Fund	Axis	35.11%	0.74	26.05%	-20.96%	77.0
SBI Bluechip Fund	SBI	30.46%	0.92	23.18%	-15.01%	74.2
Mirae Asset Tax Saver Fund	Mirae	29.18%	0.97	28.25%	-16.40%	73.7

4.1 Risk and Downside Highlights

• **Alpha Performance:** Mirae Asset Large Cap Fund leads the group with an annualized alpha of 26.96% above Nifty 100.
• **Downside Volatility:** Mid-cap and small-cap funds demonstrated higher Sortino ratios but suffered drawdowns exceeding 18% during corrections, reinforcing the need for long-term holding strategies.

5. Advanced Risk Analytics & Investor Churn

To provide institution-level financial intelligence, we deployed three advanced modules: Cohort Analysis, Value at Risk (VaR), and Churn Prediction.

5.1 Value at Risk (VaR) & CVaR

We computed daily Historical VaR (95% confidence) and Conditional VaR (CVaR). VaR represents the worst-case expected loss on a single day, while CVaR measures the average loss in the worst 5% of trading sessions. For example, large-cap equity funds exhibit a daily VaR of around -1.25%, whereas small-cap schemes exhibit a VaR exceeding -1.80%.

5.2 Investor Cohort Analysis

We segmented investors based on their first transaction year (2024 vs 2025 cohorts) and compared behaviors. The 2024 investor cohort has a higher cumulative transaction count, while the 2025 cohort demonstrates higher average monthly SIP ticket sizes, reflecting inflation and retail participation growth.

5.3 SIP Churn Prediction (Continuity)

We implemented a systematic continuity tracker that monitors the elapsed days between consecutive SIP transactions for investors with at least 6 active cycles. Investors with transaction gaps exceeding 35 days were flagged as 'at-risk' of churning or canceling their SIP plans. This is an early warning system for AMC distribution teams.

6. Business Recommendations & Conclusions

1. **Targeted Campaign for At-Risk SIPs:** AMC marketing teams should leverage the churn warning flags to trigger automated UPI mandate reminders for investors whose transaction gap approaches 30 days, preventing drop-offs.

2. **Expand B30 Geographical Outlets:** Retail transaction volumes indicate a high concentration in primary states (Punjab, TN, Rajasthan). Deploying distributor hubs in Beyond-30 (B30) tier cities presents massive untapped growth vectors.

3. **Promote Low-Expense Direct Funds:** Slicing funds by expense ratios shows a clear cost advantage for direct schemes. Advisories should highlight these products to self-service retail cohorts to drive platform loyalty.

4. **Dynamic Risk Profiling:** Integrate the custom recommender.py logic directly into client portals, matching low-risk, moderate-risk, and high-risk appetites to objective Sharpe rankings.